

AED — Adaptive Ecosystem Development

Master Development Framework v3

AED is a continuous ecosystem-development methodology that combines structured development cycles with a persistent knowledge layer, enabling humans and AI agents to design, build, activate, observe, learn from, and continuously evolve living digital ecosystems.

0. AED FOUNDATION

Before the cycles, AED has four permanent principles.

Principle 1 — Ecosystem First

Do not begin with screens or code.

Understand:

People
+ Behavior
+ Value
+ Information
+ Trust
+ Economics
+ Technology
+ Intelligence

Principle 2 — Experience Before Implementation

Technical architecture is not enough.

AED separates:

Architecture
↓
Experience Architecture
↓

Build

The system must work technically **and** make sense experientially.

Principle 3 — Activation Before Shadow

A built product is not necessarily a functioning ecosystem.

Therefore:

Build

↓

System Activation

↓

Shadow

Real data, real interactions, and real system behavior must exist before meaningful reality-vs-simulation testing can occur.

Principle 4 — Persistent Knowledge

AED never intentionally starts from zero.

Capture

↓

Organize

↓

Validate

↓

Apply

↓

Test

↓

Learn

↓

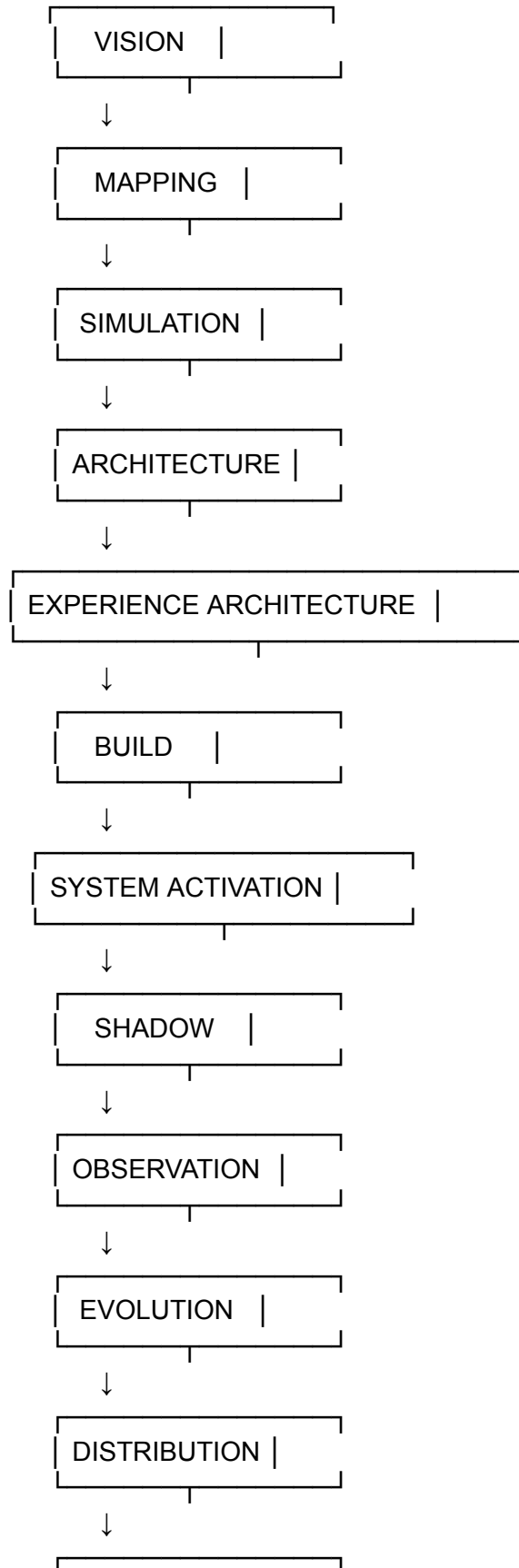
Update

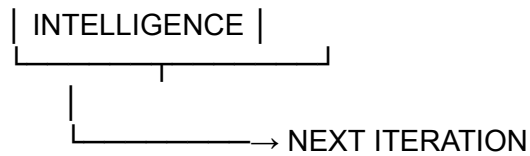
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Reuse

The Knowledge Base is therefore a **cross-cutting layer**, not another development phase.

AED MASTER FLOW

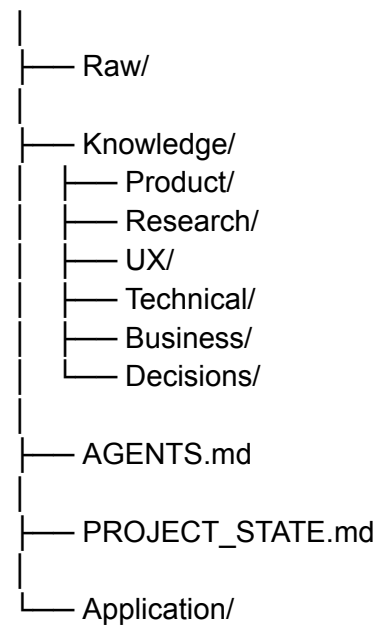




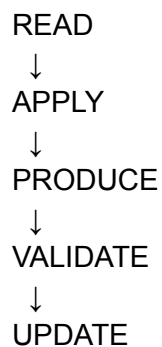
CROSS-CUTTING KNOWLEDGE LAYER

The entire flow operates over:

PROJECT/



Every cycle can:



CYCLE 1 — VISION

Purpose

Define the future ecosystem before deciding how to build it.

Vision is about **direction**, not implementation.

Core question

What are we trying to bring into existence?

Explore

- Problem
- Opportunity
- Users
- Desired transformation
- Ecosystem purpose
- Long-term ambition
- Product identity
- Core values
- Strategic boundaries

Questions

- Why should this exist?
- What problem is important enough to solve?
- Who experiences the problem?
- What changes if we succeed?
- What should users be able to do?
- What should the ecosystem become?
- What should it never become?

Knowledge READ

Knowledge/Product/

Knowledge/Research/

Knowledge/Business/

Knowledge/Decisions/

Outputs

- Vision
- Mission
- Problem definition
- Opportunity definition
- Target users
- Product philosophy

- Core principles
- Strategic assumptions

Knowledge UPDATE

Update:

Knowledge/Product/
Knowledge/Business/
Knowledge/Decisions/

Gate

Do not proceed until:

The team can clearly explain **why the ecosystem should exist and what transformation it is designed to create.**

CYCLE 2 — ECOSYSTEM MAPPING

Purpose

Understand the ecosystem as a network rather than a collection of features.

Core question

What exists inside and around the system, and how does everything interact?

Map

Actors

- Users
- Businesses
- Administrators
- Partners
- AI agents
- External systems

Relationships

Who interacts with whom?

Value

What does each actor give and receive?

Information

What information moves through the system?

Trust

How is trust created, measured and lost?

Economics

Where does money or economic value move?

Content

How is content created, discovered and distributed?

Intelligence

What does the system know?

Governance

Who can make decisions?

Outputs

- Ecosystem map
- Actor map
- Relationship map
- Value-flow map
- Information-flow map
- Trust-flow map
- Economic-flow map
- System boundaries

Knowledge READ

Knowledge/Product/
Knowledge/Research/
Knowledge/Business/

Knowledge UPDATE

Knowledge/Product/

Knowledge/Business/
Knowledge/Technical/

Gate

You should be able to answer:

"If this ecosystem were a living organism, what are its major organs and how do they interact?"

CYCLE 3 — SIMULATION

Purpose

Explore how the proposed ecosystem could behave before committing heavily to implementation.

Core question

What happens when this system meets reality?

Simulate multiple dimensions.

3.1 User Simulation

Test:

- onboarding
 - discovery
 - adoption
 - engagement
 - abandonment
 - retention
 - confusion
 - misuse
-

3.2 Economic Simulation

Test:

- pricing
 - incentives
 - transaction flows
 - marketplace liquidity
 - revenue
 - costs
 - fraud
 - payment failure
-

3.3 Trust Simulation

Test:

- fake reviews
 - reputation manipulation
 - impersonation
 - disputes
 - bad actors
-

3.4 Technical Simulation

Test:

- scaling
 - failure
 - latency
 - dependency failure
 - data loss
 - security threats
-

3.5 AI Simulation

Test:

- hallucinations
- incorrect recommendations
- bad context
- bias
- prompt manipulation

- unsafe behavior
-

3.6 Distribution Simulation

Test:

- sharing
- referrals
- virality
- spam
- network effects

Outputs

- Scenarios
- Hypotheses
- Risk register
- Failure scenarios
- Mitigation strategies
- Assumptions requiring validation

Knowledge READ

Research + existing decisions + previous lessons.

Knowledge UPDATE

Record:

Knowledge/Research/
Knowledge/Decisions/

Gate

Before Architecture:

Known high-impact risks and assumptions must be identified.

CYCLE 4 — ARCHITECTURE

Purpose

Translate the ecosystem model into a technical system.

Core question

What must exist technically for the ecosystem to function?

Define

Application Architecture

- frontend
- backend
- APIs
- services

Data Architecture

- entities
- relationships
- schemas
- storage
- indexing

Infrastructure

- hosting
- deployment
- networking
- environments

AI Architecture

- models
- orchestration
- context
- retrieval
- tools
- agents

Security

- authentication
- authorization
- encryption
- secrets

- audit

Integration

- payments
- notifications
- external APIs
- third-party systems

Technical Constraints

- budget
- performance
- scalability
- compatibility
- technology decisions

Knowledge READ

Product

Research

Technical

Decisions

Knowledge UPDATE

Knowledge/Technical/

Knowledge/Decisions/

Gate

Architecture must satisfy:

Vision

+

Ecosystem requirements

+

Simulation findings

+

Experience requirements

CYCLE 5 — EXPERIENCE ARCHITECTURE

Purpose

Define how humans experience the system.

This is one of the defining AED stages.

Core question

What should it feel like to use this ecosystem?

5.1 Product Personality

Define:

- personality
 - tone
 - behavior
 - emotional character
-

5.2 Mental Model

What should users think the product *is*?

For example:

Personal OS could be:

Companion

+

Memory

+

Thinking environment

rather than:

Dashboard

+

Analytics

+

Settings

5.3 Entry Experience

What happens when the user opens it?

5.4 Primary Action

What is the most important action?

5.5 Navigation

How does the user mentally navigate the ecosystem?

5.6 User Journey

Define:

First Contact

↓

Onboarding

↓

First Value

↓

Regular Use

↓

Habit

↓

Long-Term Relationship

5.7 Information Architecture

What information appears where?

5.8 Interaction Architecture

How do users interact with:

- content
- AI
- other users
- transactions
- knowledge
- notifications?

5.9 Emotional Architecture

What should users feel at different points?

5.10 UI/UX Architecture

Only now do we formalize:

- screens
- components
- layouts
- design system
- accessibility
- responsive behavior

Knowledge READ

Product
Research
UX
Decisions

Knowledge UPDATE

Knowledge/UX/
Knowledge/Product/
Knowledge/Decisions/

Gate

Before Build:

The team should understand what the user is supposed to experience—not merely what screens need to be created.

CYCLE 6 — BUILD

Purpose

Turn architecture and experience into functioning software.

Core question

Can we construct what we designed?

Build Areas

- Frontend
- Backend
- Database
- APIs
- Authentication
- Authorization
- AI
- Storage
- Payments
- Notifications
- Integrations
- Infrastructure

AED Build Rule

Specification

↓

Architecture

↓

Implementation

Not:

AI guesses

↓
Code

AI coding protocol

Before significant implementation:

AGENTS.md

↓

PROJECT_STATE.md

↓

Relevant Knowledge

↓

Existing Code

↓

Implementation

Knowledge READ

Technical + UX + Product + Decisions.

Knowledge UPDATE

Important implementation decisions go into:

Knowledge/Technical/
Knowledge/Decisions/

Gate

The implementation must be:

- functional
- consistent with specification
- consistent with architecture
- consistent with UX decisions

CYCLE 7 — SYSTEM ACTIVATION

Purpose

Turn the built product into a **living system**.

Core question

Can the ecosystem operate using real data and real interactions?

Stage 7.1 — Infrastructure Activation

Verify:

- database
 - authentication
 - storage
 - APIs
 - permissions
 - deployment
-

Stage 7.2 — Real User Activation

Create actual users.

Not:

John Doe
test@test.com

unless that's explicitly a test account.

Use real operational data where appropriate.

Stage 7.3 — Real Data Activation

Verify:

- creation
- storage
- retrieval
- modification
- relationships
- deletion rules

Stage 7.4 — AI Context Activation

Can AI access:

- user data?
 - history?
 - preferences?
 - goals?
 - knowledge?
 - relevant context?
-

Stage 7.5 — Retrieval Activation

Test:

Capture

↓

Index

↓

Retrieve

↓

Correct context

Stage 7.6 — Insight Activation

Can the system generate meaningful insights from real data?

Stage 7.7 — Longitudinal Activation

Can the system understand:

Today

↓

This week

↓

This month

↓

Long-term change

Output

Activated Ecosystem

Gate

Do not enter Shadow until:

Real Data

+

Real Interaction

+

Working Retrieval

+

Working Intelligence

exist.

CYCLE 8 — SHADOW

Purpose

Compare simulation with actual reality.

Core question

How different is reality from what we expected?

Compare:

Predicted Behavior

VS

Observed Behavior

Observe

- user behavior
- system behavior
- AI behavior
- economic behavior

- trust behavior
- performance
- unexpected outcomes

Important artifact

Simulation Gap Report

Prediction

Reality

Difference

Cause

Impact

Response

Knowledge UPDATE

Lessons go into appropriate domains:

Knowledge/UX/

Knowledge/Technical/

Knowledge/Product/

Knowledge/Business/

Knowledge/Decisions/

Gate

The objective isn't "everything matched."

The objective is:

Understand where reality disagrees with our assumptions.

CYCLE 9 — OBSERVATION

Purpose

Continuously understand ecosystem behavior.

Shadow asks:

"What differs from our prediction?"

Observation asks:

"What is actually happening?"

Observe

User Health

- activation
- engagement
- retention
- churn

Experience Health

- friction
- confusion
- satisfaction

Technical Health

- errors
- performance
- availability

Economic Health

- transactions
- revenue
- costs
- liquidity

Trust Health

- disputes
- fraud
- reputation

AI Health

- accuracy
- usefulness
- contextual quality

Distribution Health

- referrals
- shares
- organic acquisition
- network effects

Output

Ecosystem Health Model

CYCLE 10 — EVOLUTION

Purpose

Use evidence to decide what the ecosystem should become next.

Core question

What needs to change for the ecosystem to become healthier, more useful, and more adaptive?

Possible actions:

Improve
Remove
Simplify
Expand
Refactor
Rebalance
Automate
Redesign
Experiment

Decision process

Observation

↓

Problem / Opportunity

↓

Evidence

↓

Possible Solutions

↓

Simulation



Decision



Implementation

Knowledge UPDATE

Update:

- Product
- UX
- Technical
- Business
- Decisions

Gate

Every significant evolution should have an identifiable reason.

Avoid:

"Let's change it because this seems better."

Prefer:

"Evidence X indicates problem Y, therefore change Z."

CYCLE 11 — DISTRIBUTION

Purpose

Design how the ecosystem naturally spreads.

Core principle

Distribution is part of product architecture, not merely post-development marketing.

External Distribution

- referrals

- social sharing
 - creator networks
 - communities
 - partnerships
 - network effects
-

Internal / In-App Distribution

Behavioral Marketing

Usage itself creates visibility.

Identity Marketing

Users' profiles demonstrate reputation.

Milestone Marketing

Achievements become shareable moments.

Success Marketing

Successful outcomes become stories.

Content Marketing

User-generated content becomes discovery.

System-Generated Artifacts

Examples:

- achievement cards
 - booking completion
 - before/after results
 - progress reports
 - milestones
 - reputation achievements
-

Distribution Test

For every major feature ask:

Can this naturally generate discovery, trust, sharing, or acquisition?

Not:

"How do we force users to advertise?"

Knowledge READ

Product

UX

Business

Research

Knowledge UPDATE

Capture what actually creates organic growth.

CYCLE 12 — INTELLIGENCE

Purpose

Make the ecosystem increasingly capable of understanding itself.

Core question

What has the ecosystem learned, and what should it do with that knowledge?

Intelligence Sources

Current Data

+

Historical Data

+

Knowledge Base

+

User Behavior

+

Research

+

Decisions

+

Lessons

+

Experiments

Intelligence Functions

Pattern Detection

What repeatedly happens?

Prediction

What is likely to happen?

Recommendation

What should happen next?

Anomaly Detection

What looks unusual?

Risk Detection

What might go wrong?

Opportunity Detection

Where is there potential?

Personalization

What should this particular user experience?

THE KNOWLEDGE LAYER

The twelve cycles continuously interact with:

Raw/

Raw means:

What we encountered.

And:

Knowledge/

Knowledge means:

What we currently understand and consider useful.

KNOWLEDGE DOMAINS

Product

Contains:

- vision
 - product concepts
 - requirements
 - ecosystem rules
 - product principles
-

Research

Contains:

- market research
 - technical research
 - competitor research
 - user research
 - academic research
 - evidence
-

UX

Contains:

- experience architecture

- personas
 - journeys
 - UI decisions
 - interaction patterns
 - design principles
-

Technical

Contains:

- architecture
 - database decisions
 - API architecture
 - infrastructure
 - AI architecture
 - engineering constraints
-

Business

Contains:

- business model
 - pricing
 - strategy
 - market
 - partnerships
 - growth
-

Decisions

Contains:

- important decisions
 - rationale
 - status
 - affected systems
 - superseded decisions
 - unresolved decisions
-

KNOWLEDGE PROTOCOL

Every AED cycle uses:

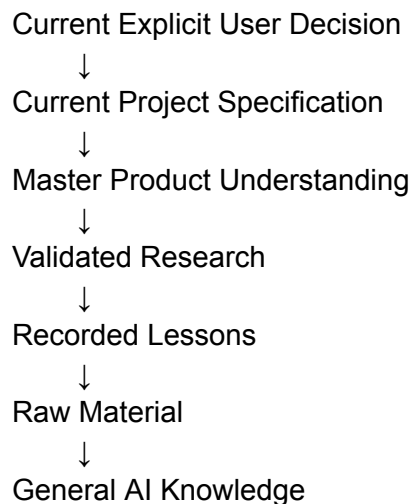
1. READ
2. CAPTURE
3. SYNTHESIZE
4. APPLY
5. VALIDATE
6. LEARN
7. UPDATE

Not every task needs all seven explicitly.

But significant work should follow the pattern.

KNOWLEDGE AUTHORITY

Use:



But contradictions are **surfaced**, not silently discarded.

CONFLICT PROTOCOL

When the AI finds:

Old Decision
VS
New Evidence

it should:

Identify conflict
↓
Present evidence
↓
Determine impact
↓
Ask/decide whether change is required
↓
Record decision
↓
Update affected knowledge

Never silently rewrite history.

DECISION MEMORY

Every consequential decision should ideally contain:

Decision:
Why:
Evidence:
Status:
Date:
Affected Systems:
Supersedes:
Consequences:

This gives future AI sessions the missing "why."

KNOWLEDGE LIFECYCLE

RAW

↓

INGESTED
↓
PROCESSED
↓
CLASSIFIED
↓
VALIDATED
↓
CANONICALIZED
↓
APPLIED
↓
TESTED
↓
LEARNED
↓
UPDATED

Not every piece of information needs to reach the final stage.

Some things remain raw because they aren't important enough to promote.

AGENTS.MD ROLE

AGENTS.md is the **AI operating manual**.

It should tell an AI agent:

1. What AED is.
 2. What cycle is currently active.
 3. Where knowledge lives.
 4. What files should be read.
 5. What decisions are binding.
 6. What constraints exist.
 7. How conflicts are handled.
 8. How new knowledge is recorded.
 9. When not to create documentation.
 10. How to update project state.
-

PROJECT_STATE.MD

I strongly recommend keeping this small root-level file.

It answers:

Where are we right now?

Example:

Current AED Cycle:
System Activation

Completed:
Vision
Mapping
Simulation
Architecture
Experience Architecture
Build

Current Objectives:
- Activate real database
- Connect authentication
- Load real data
- Validate retrieval
- Validate AI context

Blockers:
...

Next Cycle:
Shadow

This works together with [AGENTS.md](#).

AGENTS.md
= How should I operate?

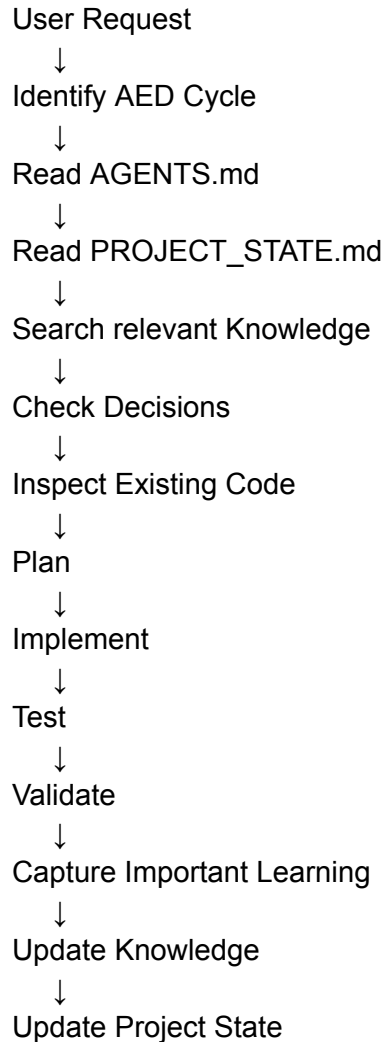
PROJECT_STATE.md
= Where are we?

Knowledge/
= What do we know?

Raw/
= What have we encountered?

THE AI SESSION LOOP

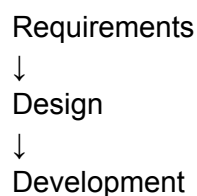
This is how an AI coding session should ideally work:



That is what makes AED particularly suited to **AI-assisted/vibe development**.

AED'S REAL DIFFERENCE

Traditional SDLC tends to look like:



↓
Testing
↓
Deployment
↓
Maintenance

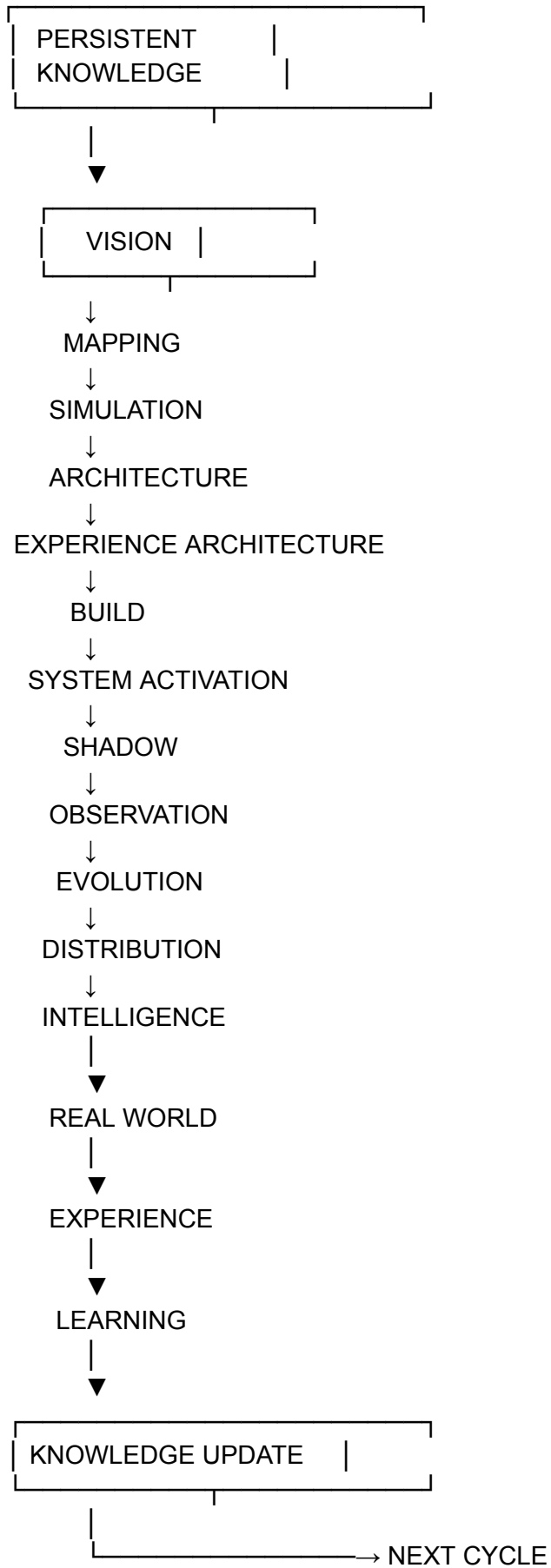
AED is more like:

Vision
↓
Ecosystem
↓
Simulation
↓
Architecture
↓
Experience
↓
Build
↓
Activation
↓
Reality
↓
Observation
↓
Evolution
↓
Distribution
↓
Intelligence
↓
Knowledge
↓
Next Reality

The difference is that AED treats the product as a **living ecosystem** rather than a static software artifact.

THE COMPLETE AED META-LOOP

Ultimately, the framework operates like this:



The fundamental AED equation

AED = Ecosystem Development + Experience Design + Real-World Activation + Continuous Learning + Persistent Knowledge

And the key idea behind the entire methodology is:

**Build the ecosystem. Activate it. Observe reality. Learn from reality.
Preserve that learning. Then use the accumulated knowledge to build the
next, better version of the ecosystem.**

That is the version I would now treat as the **canonical AED framework** going forward.